NETWORK SCANNING AND SECURITY ANALYSIS USING NMAP

A Technical Guide To Network Discovery And Security Auditing

ABSTRACT

This paper presents an overview of the functionality and operational mechanisms of Nmap as a network scanning and reconnaissance tool. Nmap operates by transmitting specially crafted packets to target systems and analyzing their responses to infer the status of hosts and services. The tool supports a variety of scanning techniques, including TCP SYN scanning, TCP connect scanning, UDP scanning, and stealth scanning methods, allowing efficient network exploration while minimizing detection.

Furthermore, Nmap integrates advanced features such as service version detection, network topology mapping, and the Nmap Scripting Engine (NSE), which enables automated vulnerability detection and customized security assessments. These capabilities make Nmap an essential utility for penetration testing, vulnerability analysis, and network inventory management.

The study highlights the importance of network scanning tools like Nmap in identifying security weaknesses, evaluating firewall configurations, and improving the overall security posture of an organization's network infrastructure. Through systematic scanning and analysis, Nmap assists in proactive defense strategies and effective network monitoring.

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Introduction

In modern computer networks, maintaining security and visibility of connected devices is essential for protecting digital infrastructure. Network administrators and cybersecurity professionals must regularly monitor networks to identify active hosts, open ports, and running services that may expose potential vulnerabilities. Network scanning tools play a critical role in this process by enabling systematic analysis of network devices and communication ports.

One of the most widely used tools for network discovery and security auditing is Nmap. Nmap, which stands for **Network Mapper**, is an open-source utility designed to explore networks, identify active hosts, and detect services running on remote systems. It was developed by Gordon Lyon (also known as Fyodor) and has become an industry-standard tool used by network administrators, ethical hackers, and security researchers.

Nmap operates by sending specially crafted packets to target hosts and analyzing the responses received from those systems. Through this process, it can determine which devices are online, which ports are open or closed, what services are running on those ports, and in many cases the operating system of the target device. These capabilities make Nmap a powerful tool for network auditing, vulnerability assessment, and system management.

In addition to basic network discovery, Nmap provides advanced features such as service version detection, operating system fingerprinting, and automated scripting through its scripting engine. These features allow security professionals to perform detailed security assessments and identify potential weaknesses within a network environment.

Objectives

The objective of this documentation is to provide a clear understanding of network scanning concepts and the practical usage of Nmap in network security analysis. Network scanning plays an important role in identifying devices, services, and potential vulnerabilities within a computer network. By studying Nmap and its functionalities, users can gain practical knowledge of how network reconnaissance and security auditing are performed.

The specific objectives of this documentation are as follows:

1. To understand the concept of network scanning and its importance in maintaining network security.
2. To study the architecture and working principles of Nmap, including how it sends packets and analyzes responses from target systems.
3. To learn the command structure of Nmap, including the different options and parameters used to perform network scans.
4. To explore various scanning techniques, such as TCP SYN scan, TCP connect scan, and UDP scan.
5. To understand firewall and intrusion detection system (IDS) evasion techniques used during network scanning and security testing.
6. To analyze scan results, including identifying open, closed, and filtered ports on a target system.
7. To demonstrate practical examples of Nmap scans and interpret the results for security assessment.

By achieving these objectives, this documentation aims to provide both theoretical knowledge and practical insights into how network scanning tools are used by network administrators and cybersecurity professionals to monitor and secure computer networks.

3. Overview of Nmap

Nmap (Network Mapper) is an open-source tool used for network discovery and security auditing. It allows administrators and security professionals to analyze networks by identifying active hosts, detecting open ports, discovering running services, and determining operating systems of devices connected to a network.

Nmap works by sending specially crafted packets to target systems and analyzing the responses it receives. Based on these responses, it can determine the status of network ports and identify services running on them. Because of its flexibility and powerful scanning capabilities, Nmap has become one of the most widely used tools in network administration, penetration testing, and cybersecurity research.

3.1 Definition and Purpose

Nmap is a network scanning utility designed to explore networks and identify devices connected to them. It helps users collect detailed information about network systems and their services.

The primary purposes of Nmap include:

- Network Discovery – Identifying active hosts within a network.
- Port Scanning – Detecting open, closed, or filtered ports on target devices.
- Service Detection – Determining the applications and services running on open ports.
- Operating System Identification – Detecting the operating system of a target host.
- Security Auditing – Identifying potential vulnerabilities and misconfigurations in a network.

By providing these capabilities, Nmap helps organizations monitor their network infrastructure and improve overall security.

3.2 History and Development

Nmap was originally developed in 1997 by Gordon Lyon, who is also known by the pseudonym *Fyodor*. The tool was first introduced as a small utility for scanning network ports but quickly gained popularity due to its effectiveness and reliability.

Over time, Nmap evolved into a powerful and comprehensive network scanning platform. Continuous development by the global open-source community has added many advanced features such as service detection, operating system fingerprinting, and scripting capabilities.

Today, Nmap is considered an industry standard tool used by network administrators, security analysts, and ethical hackers worldwide.

3.3 Key Features

Nmap provides a wide range of features that make it a powerful tool for network analysis and security assessment.

1. Host Discovery

Nmap can identify active devices within a network by sending various types of probes such as ICMP requests, TCP packets, and ARP requests.

2. Port Scanning

It can scan thousands of ports to determine whether they are open, closed, or filtered by a firewall.

3. Service and Version Detection

Nmap can identify services running on open ports and determine the software versions used by those services.

4. Operating System Detection

Through TCP/IP fingerprinting techniques, Nmap can estimate the operating system running on a target device.

5. Nmap Scripting Engine (NSE)

Nmap includes a powerful scripting engine that allows users to run scripts for vulnerability detection, network discovery, and service enumeration.

6. Flexible Command-Line Interface

Nmap provides numerous command-line options that allow users to customize scans according to their specific requirements.

These features make Nmap an essential tool for performing detailed network reconnaissance and security analysis.

4. Nmap Command Structure

The command structure of Nmap allows users to perform different types of network scans using a variety of parameters and options. Nmap is primarily operated through a command-line interface where users specify the target system and the scanning methods they want to use.

The flexibility of Nmap's command structure enables users to perform simple scans to detect open ports or advanced scans that include service detection, operating system identification, and vulnerability analysis. Understanding how Nmap commands are structured is essential for performing efficient and accurate network reconnaissance.

4.1 Basic Syntax of Nmap

The general syntax of an Nmap command is as follows:

Command syntax

```
nmap [Scan Type] [Options] [Target]
```

Each part of this command has a specific purpose:

- **Scan Type** determines the method used to scan ports on the target system.
- **Options** define additional parameters such as port ranges, service detection, or output format.
- **Target** specifies the host or network that will be scanned.

Example command:

```
nmap -sS 192.168.1.1
```

Explanation:

Component	Description
nmap	Executes the Nmap scanning tool
-sS	Performs a TCP SYN scan
192.168.1.1	Target IP address

This command performs a basic SYN scan on the specified host.

4.2 Target Specification

Target specification defines which systems or networks will be scanned. Nmap supports several methods for specifying targets, allowing users to scan individual hosts, ranges of IP addresses, or entire network segments.

Examples of target specification:

1. **Single Host:**

```
nmap 192.168.1.10
```

This command scans one specific device.

2. **Multiple Hosts:**

```
nmap 192.168.1.10 192.168.1.15
```

This command scans multiple devices.

3. **IP Range:**

```
nmap 192.168.1.1-50
```

This scans all hosts from IP address 192.168.1.1 to 192.168.1.50.

4. **Subnet Scan:**

```
nmap 192.168.1.0/24
```

This scans all devices in the subnet.

5. **Domain Name**

nmap example.com

This scans the server associated with the domain.

These target specification options provide flexibility when scanning different network environments.

4.3 Command-Line Options

Nmap includes many command-line options that allow users to control how scans are performed. These options determine scanning techniques, speed, detection methods, and output formats.

Some commonly used command-line options include:

Option	Description
-sS	Performs a TCP SYN scan
-sT	Performs a TCP connect scan
-sU	Performs a UDP scan
-p	Specifies ports to scan
-sV	Detects service versions
-O	Detects operating system
-A	Performs aggressive scan with multiple detection techniques
-T0 to -T5	Controls scan speed and timing

Example command using multiple options:

nmap -sS -sV -O -p 80 192.168.1.1

Explanation:

- -sS performs a SYN scan
- -sV detects the service version
- -O identifies the operating system
- -p 80 scans port 80 on the target host

These command-line options allow users to customize scans according to their requirements and perform detailed network analysis.

5. Scanning Techniques

Scanning techniques are the core methods used by Nmap to gather information about target systems in a network. These techniques determine how Nmap communicates with the target host and how it interprets the responses received from network ports.

Different scanning techniques provide different advantages in terms of **speed, stealth, accuracy, and reliability**. Some scans are designed to be fast and efficient, while others aim to avoid detection by firewalls and intrusion detection systems. Understanding these techniques helps network administrators and security professionals choose the most appropriate scanning method for a specific task.

5.1 TCP SYN Scan

The **TCP SYN scan**, also known as a **half-open scan**, is one of the most commonly used scanning techniques in Nmap. It is often referred to as a ***stealth scan*** because it does not complete the full TCP connection process.

In a normal TCP connection, a **three-way handshake** occurs between the client and the server:

1. SYN (synchronization request) sent by the client
2. SYN-ACK (acknowledgment) returned by the server
3. ACK (acknowledgment) sent by the client to establish the connection

During a SYN scan, Nmap sends a ***SYN packet*** to the target port. The response from the target determines the state of the port:

Response	Port State
SYN-ACK	Port is **Open**
RST	Port is **Closed**
No Response / ICMP error	Port is **Filtered**

Since the connection is never fully established, SYN scans are faster and less likely to be logged by some systems.

Example command:

nmap -sS 192.168.1.10

5.2 TCP Connect Scan

The **TCP Connect scan** performs a complete TCP three-way handshake with the target system. Unlike the SYN scan, this technique fully establishes a connection with the target port before closing it.

This scan is typically used when the user does not have the administrative privileges required to perform a SYN scan.

The process follows these steps:

1. Nmap sends a SYN packet.
2. The target responds with SYN-ACK if the port is open.
3. Nmap sends an ACK packet to complete the connection.
4. The connection is then closed.

Because the full connection is established, this scan is more easily detected by security monitoring systems.

Example command:

nmap -sT 192.168.1.10

5.3 UDP Scan

The **UDP scan** is used to identify open **UDP ports** on a target system. Unlike TCP, the UDP protocol does not use a connection-based handshake process, which makes scanning more complex.

When Nmap sends a UDP packet to a port:

Response	**Port State**
UDP response received	Port is **Open**
ICMP Port Unreachable	Port is **Closed**
No response	Port may be **Open or Filtered**

UDP scans are typically slower than TCP scans because the lack of response makes it difficult to determine the exact port state.

Example command:

nmap -sU 192.168.1.10

UDP scans are often used to detect services such as:

- DNS (Port 53)

- SNMP (Port 161)
- DHCP (Ports 67 and 68)

5.4 FIN, NULL, and Xmas Scans

These scans are advanced techniques that manipulate **TCP flag settings** to identify open ports and potentially bypass certain firewall filters.

FIN Scan

In a FIN scan, Nmap sends a packet with the **FIN flag** set. According to TCP protocol standards:

- Closed ports respond with **RST packets**.
- Open ports usually **ignore the packet**.

Example command:

```
nmap -sF 192.168.1.10
```

NULL Scan

A NULL scan sends packets with **no TCP flags set**.

Expected behavior:

- Closed ports respond with **RST packets**.
- Open ports typically **do not respond**.

Example command:

```
nmap -sN 192.168.1.10
```

Xmas Scan

The Xmas scan sets multiple TCP flags simultaneously, typically **FIN, PSH, and URG**. The packet appears “lit up” with flags, similar to a Christmas tree.

Responses are interpreted similarly:

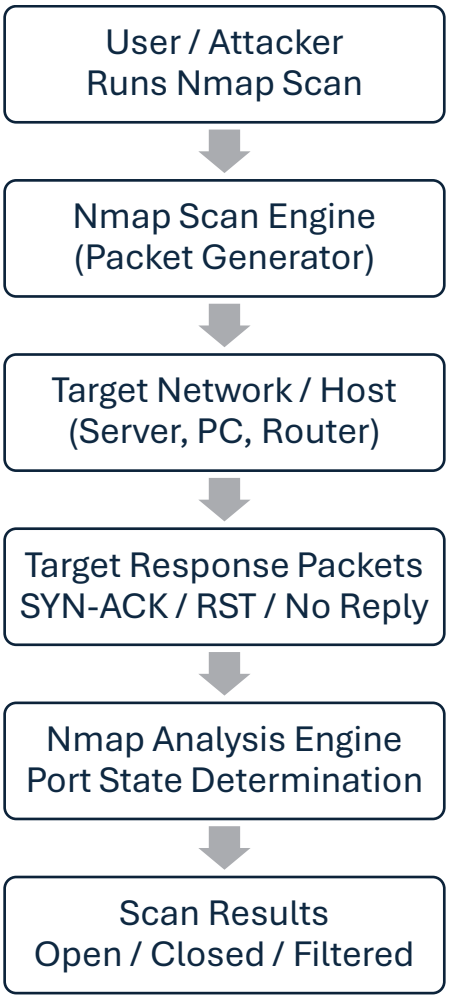
- Closed ports respond with **RST packets**.
- Open ports usually **ignore the packet**.

Example command:

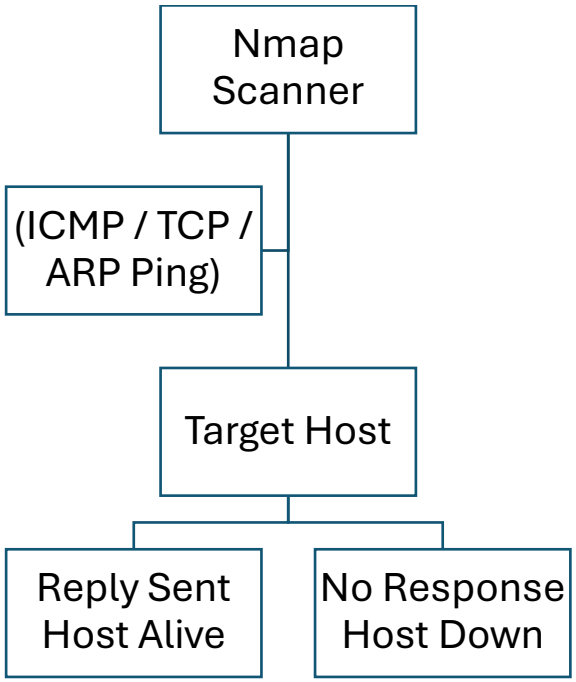
nmap -sX 192.168.1.10

These scanning techniques allow Nmap to gather detailed information about network systems while adapting to different security environments. By selecting the appropriate scanning method, users can efficiently analyze network configurations and identify potential security issues.

Basic Nmap Scanning Process



Host Discovery Process



Full Nmap Scanning Workflow

6. Firewall / IDS Evasion Methods

Firewalls and Intrusion Detection Systems (IDS) are security mechanisms designed to monitor and control network traffic. They help detect suspicious activities such as unauthorized network scans or potential attacks. However, advanced network scanning tools like Nmap provide several techniques that can reduce the chances of detection by these security systems.

Firewall and IDS evasion methods work by modifying the structure, origin, or characteristics of network packets sent during scanning. These techniques are mainly used in **penetration testing and security assessments** to evaluate how well a network's defense mechanisms can detect and block scanning attempts.

The following subsections describe some common evasion methods available in Nmap.

6.1 Packet Fragmentation

Packet fragmentation is a technique where large packets are divided into smaller fragments before being transmitted to the target system. Some firewalls and IDS devices analyze packets individually and may fail to correctly reassemble fragmented packets for inspection.

By splitting packets into smaller fragments, the scanning tool may bypass certain filtering rules that rely on inspecting complete packets.

Example command:

nmap -f 192.168.1.1

Explanation:

Option	Function
-f	Fragment packets into smaller pieces to evade firewall detection

This technique may help bypass older firewall systems that cannot properly reconstruct fragmented packets.

6.2 Decoy Scanning

Decoy scanning is used to hide the real source of a scan by generating traffic from multiple fake IP addresses along with the attacker's real IP address. This makes it difficult for intrusion detection systems to determine which system is actually performing the scan.

In this method, the target system receives scan packets appearing to originate from several different hosts.

Example command:

nmap -D RND:5 192.168.1.1

Explanation:

Option	Function
-D	Enables decoy scanning
RND:5	Generates 5 random decoy IP addresses

This technique can confuse monitoring systems and make tracking the true scanning source more difficult.

6.3 Source Port Manipulation

Some firewall rules allow traffic only from specific trusted ports such as DNS or HTTP ports. Nmap can manipulate the source port number of scanning packets to appear as if the traffic is coming from a trusted service.

By using commonly allowed ports, the scan may bypass firewall filtering rules.

Example command:

nmap --source-port 53 192.168.1.1

Explanation:

Option	Function
--source-port	Specifies a custom source port for packets
53	Common DNS port often allowed through firewalls

This method attempts to make scanning traffic appear legitimate.

6.4 Random Data Padding

Random data padding involves adding random bytes to scanning packets. This increases the packet size and alters its structure, which may prevent simple signature-based IDS systems from recognizing the scan pattern.

By changing the packet structure, the scan becomes harder to detect using pattern matching.

Example command:

nmap --data-length 50 192.168.1.1

Explanation:

Option	Function
--data-length	Adds random data to packets
50	Specifies the number of random bytes to append

This technique can help avoid detection by some intrusion detection systems that rely on fixed packet signatures.

These firewall and IDS evasion techniques demonstrate how flexible and powerful Nmap is for advanced network analysis and penetration testing. By using these methods, security professionals can evaluate the effectiveness of firewall rules and intrusion detection mechanisms within a network environment.

1. ***Basic Port Scan Results Table:***

   This table represents results from a simple scan:

   Command example:

```
nmap 192.168.1.10
```

Port	Protocol	State	Service	Description
22	TCP	Open	SSH	Secure remote login service
80	TCP	Open	HTTP	Web server running
443	TCP	Open	HTTPS	Secure web service
3306	TCP	Closed	MySQL	Database service not accessible
8080	TCP	Filtered	HTTP-Proxy	Possibly blocked by firewall

2.Service Version Detection Results

Command example:

```
nmap -sV 192.168.1.10
```

Port	Service	Version	Status
22/tcp	SSH	OpenSSH 8.2p1	Running
80/tcp	HTTP	Apache 2.4.41	Running
443/tcp	HTTPS	Apache SSL	Running
21/tcp	FTP	vsftpd 3.0.3	Running

Purpose of this scan:

- Identify software versions
- Detect potential vulnerabilities

3.Operating System Detection Results

Command example:

```
nmap -O 192.168.1.10
```

Target IP	Device Type	Operating System	Accuracy
192.168.1.10	Server	Linux Kernel 5.x	95%

Target IP	Device Type	Operating System	Accuracy
192.168.1.20	Router	Embedded Linux	90%
192.168.1.25	Workstation	Windows 10	92%

4. Network Host Discovery Results

Command example:

```
nmap -sn 192.168.1.0/24
```

Host IP	Hostname	Status	Latency
192.168.1.1	Router	Up	1.2 ms
192.168.1.5	Desktop-PC	Up	2.1 ms
192.168.1.10	Web-Server	Up	1.8 ms
192.168.1.25	Laptop	Down	No Response

Purpose:

- Identify **active devices in the network**

5. Full Port Scan Results

Command example:

```
nmap -p- 192.168.1.10
```

Port	State	Service
21	Open	FTP
22	Open	SSH
25	Closed	SMTP
53	Open	DNS
80	Open	HTTP
110	Closed	POP3
443	Open	HTTPS

A full scan checks all 65,535 ports.

6. Vulnerability Script Scan Results

Command example:

```
nmap --script vuln 192.168.1.10
```

Port	Service	Vulnerability Detected	Risk Level
80	HTTP	Directory Listing Enabled	Medium
21	FTP	Anonymous Login Allowed	High
445	SMB	SMBv1 Enabled	Critical

Common Ports and Their Services

Port	Protocol	Service Name	Description
20	TCP	FTP Data	File Transfer Protocol data transfer
21	TCP	FTP	File Transfer Protocol for file sharing
22	TCP	SSH	Secure remote login and command execution
23	TCP	Telnet	Remote login protocol (unencrypted)
25	TCP	SMTP	Simple Mail Transfer Protocol for sending email
53	TCP/UDP	DNS	Domain Name System for domain resolution
67	UDP	DHCP Server	Dynamic Host Configuration Protocol server
68	UDP	DHCP Client	DHCP client communication
69	UDP	TFTP	Trivial File Transfer Protocol
80	TCP	HTTP	Web server communication
110	TCP	POP3	Post Office Protocol for retrieving emails
119	TCP	NNTP	Network News Transfer Protocol
123	UDP	NTP	Network Time Protocol for clock synchronization
137	UDP	NetBIOS Name	NetBIOS name service
138	UDP	NetBIOS Datagram	NetBIOS datagram service
139	TCP	NetBIOS Session	Windows file sharing
143	TCP	IMAP	Internet Message Access Protocol

Port	Protocol	Service Name	Description
161	UDP	SNMP	Simple Network Management Protocol
179	TCP	BGP	Border Gateway Protocol
194	TCP	IRC	Internet Relay Chat
389	TCP/UDP	LDAP	Lightweight Directory Access Protocol
443	TCP	HTTPS	Secure HTTP communication
445	TCP	SMB	Server Message Block (Windows file sharing)
465	TCP	SMTPS	Secure SMTP
500	UDP	ISAKMP	VPN key exchange
514	UDP	Syslog	System logging service
515	TCP	LPD	Line Printer Daemon
520	UDP	RIP	Routing Information Protocol
587	TCP	SMTP Submission	Secure email submission
636	TCP	LDAPS	Secure LDAP
989	TCP	FTPS Data	Secure FTP data transfer
990	TCP	FTPS	Secure FTP control
993	TCP	IMAPS	Secure IMAP
995	TCP	POP3S	Secure POP3

Example Nmap Output for Ports

When running a scan with Nmap:

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

443/tcp open https

3306/tcp open mysql

Explanation:

Field	Meaning
PORT	Network port number
STATE	Open / Closed / Filtered
SERVICE	Application running on the port

7.Practical Examples

1. Basic Syntax of Nmap

General command format:

nmap [Scan Type] [Options] [Target]

Example:

nmap -sS -p 80 192.168.1.1

2. Target Specification Commands

- **Single IP**

nmap 192.168.1.1

- **Multiple IPs**

nmap 192.168.1.1 192.168.1.2

- **IP Range**

nmap 192.168.1.1-100

- **Subnet scan**

nmap 192.168.1.0/24

- **Scan a domain**

nmap example.com

- **Scan from file**

nmap -iL targets.txt

3. Host Discovery Commands

Checks **which hosts are alive**.

- **Ping Scan**

nmap -sn 192.168.1.0/24

- **Disable ping**

nmap -Pn target

Meaning: treat host as online.

- **ARP Scan**

nmap -PR target

Used for **local networks**.

4. Port Scanning Commands

- **Scan specific ports**

nmap -p 80 target

- **Scan multiple ports**

nmap -p 22,80,443 target

- **Scan port range**

nmap -p 1-1000 target

- **Scan all ports**

nmap -p- target

5. Scan Types

Determines how Nmap scans ports.

TCP SYN Scan (Stealth Scan)

nmap -sS target

Breakdown:

Option	Meaning
-sS	half-open TCP scan

- TCP Connect Scan

nmap -sT target

Used when no root privileges

- UDP Scan

nmap -sU target

Scans UDP services like DNS.

- FIN Scan

nmap -sF target

Used to bypass some firewalls.

- NULL Scan

nmap -sN target

No flags set.

- Xmas Scan

nmap -sX target

Packet flags: FIN + PSH + URG.

6. Service & Version Detection

nmap -sV target

Breakdown:

Option	**Meaning**
-sV	detect service version

Example output

80/tcp open http Apache 2.4

7. Operating System Detection

nmap -O target

Breakdown:

Option	Meaning
-O	OS detection

Example output

OS: Linux 5.x

8. Aggressive Scan

nmap -A target

Includes:

Feature	Included
OS detection	✓
Version detection	✓
Script scanning	✓
Traceroute	✓

9. Nmap Script Engine (NSE)

Run scripts for vulnerabilities.

nmap --script vuln target

Breakdown:

Option	Meaning
--script	run scripts

Example scripts

nmap --script http-enum target

nmap --script ftp-anon target

nmap --script smb-vuln target

10. Timing Options

Controls **scan speed**.

nmap -T0 target

nmap -T5 target

Breakdown:

Option	Speed
-T0	paranoid

Option	Speed
-T1	sneaky
-T2	polite
-T3	normal
-T4	fast
-T5	insane

11. Output Commands

Save scan results.

- **Normal output**

nmap -oN result.txt target

- **XML output**

nmap -oX result.xml target

- **Grepable output**

nmap -oG result.txt target

- **Save all formats**

nmap -oA result target

Breakdown:

Option	Meaning
-oN	normal
-oX	XML
-oG	grep format
-oA	all formats

12. Firewall Evasion Commands

- **Fragment packets**

nmap -f target

- **Decoy scan**

nmap -D RND:10 target

Breakdown:

Option	Meaning
-D	use decoy IPs

- **Spoof IP**

nmap -S fakeIP target

13. Verbosity Commands

Increase output detail

nmap -v target

nmap -vv target

14. Traceroute

nmap --traceroute target

Shows network path.

15. Example Real Professional Command

nmap -sS -sV -O -p- -T4 192.168.1.1

Breakdown:

Option	**Meaning**
-sS	SYN scan
-sV	version detection
-O	OS detection
-p-	scan all ports
-T4	faster scan

Conclusion

Network scanning is a fundamental process in maintaining and securing modern computer networks. It enables administrators and security professionals to identify active hosts, detect open ports, and analyze services running on network systems. Tools designed for network scanning help organizations understand their network structure and identify potential vulnerabilities before they can be exploited.

One of the most powerful and widely used tools for this purpose is Nmap. Throughout this documentation, the various aspects of Nmap have been discussed, including its command structure, scanning techniques, firewall evasion methods, and practical usage examples. These features demonstrate how Nmap can effectively perform network discovery, port scanning, service detection, and operating system identification.

Understanding different scanning techniques such as TCP SYN scans, TCP connect scans, UDP scans, and advanced methods like FIN, NULL, and Xmas scans provides deeper insight into how network communication works. Additionally, techniques for firewall and IDS evasion illustrate how security tools attempt to detect scanning activities and how penetration testers evaluate the effectiveness of these defenses.

Overall, Nmap serves as a valuable tool for network administrators, cybersecurity professionals, and ethical hackers who need to analyze and secure network environments. By using Nmap responsibly and ethically, organizations can strengthen their security posture, detect potential weaknesses, and ensure the reliability and safety of their network infrastructure.

In conclusion, mastering the concepts and techniques associated with Nmap not only improves network monitoring and troubleshooting but also contributes significantly to the field of cybersecurity and network management.

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~ Devi Kumar